



Alberta Public Health Disease Management Guidelines

Shigellosis



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Health and Wellness Promotion Branch

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Alberta Health

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Case Definition

Confirmed Case

Laboratory confirmation of infection with or without clinical illness^(A):

- Isolation of *Shigella* species from an appropriate clinical specimen (e.g., sterile site, deep tissue wounds, stool, vomit or urine)^(B)

Probable Case

Clinical illness^(A) in a person who is epidemiologically linked to a confirmed case

^(A) Clinical illness is characterized by diarrhea, fever, nausea and sometimes toxaemia, vomiting, cramps or tenesmus. Asymptomatic infections may occur.

^(B) Refer to the [Public Health Laboratory \(ProvLab\) Guide to Services](#) for current specimen collection and submission information.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases in the prescribed form by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:

- Chief Medical Officer of Health (CMOH) (or designate), and
- MOH (or designate) of the zone.

Alberta Health Services and First Nations Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- The MOH (or designate) shall report all outbreaks to CMOH (or designate) via the [Alberta Outbreak Report Form \(AORF\)](#) as soon as possible using existing processes (e.g., CDOM or fax).
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Etiology

Shigellosis is an acute bacterial disease caused by gram negative bacilli in the family Enterobacteriaceae.⁽¹⁾ It is the most communicable of the bacterial diarrheas. Some strains produce Shigatoxin type 1 (Stx1) much like the verotoxin of *E. coli* O157:H7.⁽²⁾ Approximately 40 serotypes are divided into four groups depending on serologic similarity and fermentation reactions:

- Group A - *Shigella dysenteriae*
- Group B - *Shigella flexneri*
- Group C - *Shigella boydii*
- Group D - *Shigella sonnei*

Clinical Presentation

Shigellosis is unique among bacterial enteropathogens in that a very low dose of the organism readily produces disease in humans. This may account for the high secondary attack rate in families. The severity of the presentation and case fatality rates are functions of the host.

Shigellosis involves the large and distal small intestine.⁽³⁾ Disease is most commonly characterized by diarrhea (may contain blood and mucus or be watery) accompanied by fever, nausea, and tenesmus. It may be biphasic with an initial period of watery diarrhea and cramps, followed by the development of dysentery (blood and mucus).^(1,3) Febrile convulsions may be a significant complication in young children. The illness is usually self-limited, lasting between one day and one month with an average of seven days.⁽⁴⁾ Some individuals have mild or asymptomatic infections. Bacteremia and pneumonia may occur but are uncommon.

There is a rare fulminant form of bacillary dysentery secondary to a massive small intestine invasion by the bacteria. This may be seen in children. Death early in the infection is common (the “Ikari” syndrome).⁽⁵⁾

Infection with *S. dysenteriae* is often associated with serious disease and more severe complications, including toxic megacolon and hemolytic-uremic syndrome (HUS). It is one of the most common *Shigella* species to occur in developing countries and least common in developed countries.

Some strains of *S. flexneri* can cause Reiter’s syndrome (reactive arthropathy) in persons who are genetically predisposed, although Reiter’s syndrome can occur with any of the *Shigella* strains.^(1,6–11) This is a common strain in developing countries but also accounts for approximately one quarter of cases in the United States. It is now thought to be sexually transmitted.⁽¹⁾

Shigella boydii is not a common isolate in North America but widespread in developing countries.

Infection with *S. sonnei* often results in a short clinical course. Infection may be fatal in immunocompromised individuals. This is the most common strain in developed countries. It accounts for approximately 75% of cases in the US.

Diagnosis

Diagnosis is made by the isolation of *Shigella* from feces or a rectal swab or detection of *Shigella* nucleic acid by PCR.^(1,3) The infection is generally associated with a high number of leukocytes (pus cells) in the fecal matter.

Shigella remains viable outside the human body for only a short period of time. Hence, specimens must be processed rapidly after collection, preferable within 24 hours. Use of appropriate media increases the likelihood of organism isolation. ⁽¹²⁾

Serological testing is not generally helpful.

Treatment

- Treatment is recommended for most symptomatic patients. Use of antibiotics will shorten the period of fecal excretion of the infecting strain and will shorten the clinical course of disease often to a few days and may be lifesaving in the event the case has *S. dysenteriae*.⁽⁴⁾
 - Antibiotic resistance frequently develops after treatment.
- Replace fluids and electrolytes if there is excessive fluid loss through diarrhea or vomiting.
- Antimotility agents are not recommended, as they may prolong the course of disease or play a role in the development of toxic dilation of the colon or perforation.⁽⁴⁾

Reservoir

Humans and primates are the reservoir for *shigella*.^(1,3) The source is most often the feces of infected individuals who have diarrhea. Other potential sources include contaminated food (especially when unrefrigerated, e.g., salads), raw vegetables, milk, dairy products, and water.⁽¹⁾ Outbreaks in primates such as monkeys and chimpanzees have occurred.

Transmission

The primary mode of transmission is fecal-oral from an infected person (symptomatic or asymptomatic): improper hand hygiene, lack of sanitary conditions, overcrowding, eating or drinking contaminated food or water and sexual contact are risk factors for transmitting infection.^(1,3) Direct transmission is common in children and individuals who do not thoroughly clean their hands and under the fingernails following defecation.

Incubation Period

Generally, the incubation period for *Shigella* is usually 1–3 days but can range from 12–96 hours (4 days).^(1,3) The incubation period for *S. dysenteriae* can be longer, up to one week.

Period of Communicability

Shigellosis can be spread while the person has acute infection and until the organism is no longer present in feces.^(1,3,4) This can last as long as four weeks. Long-term carriage is uncommon; however, asymptomatic carriers do have the ability to transmit disease. The duration of carriage may be reduced to a few days with the use of appropriate antibiotics.

Host Susceptibility

The individuals at highest risk of acquiring shigellosis include:

- children ≤ 5 years of age in child care settings and their caregivers,
- the elderly,
- people living in crowded conditions,
- individuals with HIV,
- men who have sex with men, and
- debilitated and malnourished individuals are at increased risk of infection.^(1–3,13)

Travellers to developing countries may be more at risk of acquiring antibiotic-resistant strains of *Shigella*.^(14–17)

Secondary attack rates can be as high as 40% in households.⁽¹⁾ Breastfeeding is protective for infants.⁽³⁾ Serotype-specific immunity can be demonstrated in those who have recovered.⁽¹⁸⁾

Incidence

Shigellosis occurs worldwide.⁽¹⁸⁾ Up to 15% of all diarrhea cases can be linked to *Shigella* infections.⁽²⁾ It is estimated that up to 165 million cases and up to 1 million deaths due to *Shigella* occur worldwide each year.^(19,20) Multi-antibiotic resistant *Shigella* strain outbreaks have appeared worldwide. Children under 5 years of age account for the highest rate of cases (69%) and the highest death rate (61%).⁽²⁰⁾ Secondary attack rates in households have been shown to be as high as 40%.⁽¹⁾ *S. dysenteriae* is more commonly reported in developing countries while *S. sonnei* is more commonly reported in developed countries.⁽¹⁾ *S. boydii* is found almost exclusively on the Indian subcontinent.⁽²¹⁾

Shigellosis is notifiable both in Alberta and Canada. The most common serotype in Alberta is *S. sonnei*. Outbreaks of shigellosis have been reported in Alberta. Sources are often difficult to identify but over-crowding is often a factor in on-going transmission. Large urban centers report the highest number of cases. Shigellosis is most prevalent in young children under the age of 10.⁽²²⁾

For more information of shigellosis cases in Alberta, refer to [Interactive Health Data Application](#).

Public Health Management

Key Investigation

- Confirm the diagnosis.
- Obtain a history of illness including the date of onset, and signs and symptoms.
- Identify any underlying medical conditions that may increase host susceptibility.
- Determine the occupation of the case and if in a **sensitive situation or occupation (SSO)** that poses a higher risk of transmission to others. See Table 1 for a definition of SSO.
- Determine the possible source of infection, taking into consideration the incubation period^(C), reservoir, and mode of transmission. Assessment may include determining, obtaining or identifying:
 - a detailed food history,
 - history of institutionalization,
 - contact with sewage contaminated recreational water,
 - consumption of untreated surface water,
 - high risk sexual practices, especially contact with feces,
 - international travel or recent immigration (especially to/from areas with inadequate sanitation, water and sewage treatment), and
 - residing in areas with poor sanitation including improper water treatment and sewage disposal.
- Assess for history of similar symptoms in other members of the household.
- Suspected contaminated food may be held or destroyed to prevent consumption.
- Identify any contacts, especially those in SSO that pose a higher risk of transmission to others – Refer to Table 1. Contacts include:
 - persons living in the household,
 - children and child care workers at a child care facility (daycare, dayhome, or other child care site), and
 - individuals exposed to the same source where the source is identified.

Table 1: Sensitive Situations or Occupations (SSO)

SSO	Activities
Food handler	• Touches unwrapped food to be consumed, <u>and/or</u> • Handles equipment or utensils that touch unwrapped food to be consumed.*
Health care, child care or other staff	• Has contact through serving food to highly susceptible patients or persons in whom an intestinal infection would have particularly serious consequences. • Provides direct patient care and is involved in the care of young children, elderly or dependent persons.
Child attending a child care facility or similar facilities	• Is diapered or unable to implement good standards of personal hygiene.
Any individual (older child or adult)	• Is unable to implement good standards of personal hygiene (e.g., with disabilities/challenges that may impact ability to perform good hand hygiene).

* NOTE: Generally, food handlers who do not touch food, equipment or utensils in this way are not considered to pose a transmission risk; however, circumstances for each case should be assessed on an individual basis.

^(C) The incubation period of shigellosis is usually 1–3 days. However, individual cases can vary, thus requiring flexibility when determining where and how the disease was likely acquired.

Management of a Case

- Antibiotics are often recommended for public health control.
- All cases should be advised of the following:
 - how the disease is transmitted, appropriate personal hygiene, routine infection prevention and control practices, and contact precautions,
 - to avoid food preparation until symptoms have resolved, and
 - to avoid public recreational water (e.g., swimming pools) until after treatment is completed and diarrhea has resolved.⁽³⁾
- Contact precautions should be used in health care settings where children or adults have poor hygiene or incontinence that cannot be contained. Otherwise, routine practices are adequate.
- Refer to Table 2 for case exclusion criteria. For a summary of all enteric exclusions refer to [Enteric Transmission Risk Assessment and Exclusion Table](#).

Table 2: Case Exclusion

Cases	Category	Exclusion Criteria
Symptomatic and Asymptomatic	SSO	• The MOH shall by order exclude the case until: – 48 hours after treatment with appropriate antibiotics has been completed, and – two stool specimens taken from the infected person are reported as negative - stools shall be collected no less than 24 hours apart and at least 48 hours after normal stools have resumed. • Reassignment to a low risk area may be used as an alternative to exclusion.
Symptomatic	Non-SSO	• No exclusion required; however, all cases of gastroenteritis or enteritis should be regarded as potentially infectious and should remain home from work, school or daycare until 48 hours after diarrhea has stopped.
Asymptomatic	Non-SSO	• No exclusion required.

Management of Contacts

- Contacts should be instructed about disease transmission, appropriate personal hygiene, routine practices, and contact precautions.
- Refer to Table 3 for contact exclusion criteria.

Table 3: Contact Exclusion

Contacts	Category	Exclusion Criteria
Symptomatic	SSO	• Same as a case. • Ensure the contact is assessed by their physician.
Asymptomatic	SSO	• The MOH may by order exclude asymptomatic contacts until one stool specimen is reported as negative. • In the case of daycare facilities, asymptomatic contacts of cases may be investigated and any children found to be positive should be cohorted where it is practical to do so. • Contacts should monitor themselves for gastrointestinal symptoms, maintain good hand hygiene and food handling practices and seek medical attention if symptoms develop.
Symptomatic	Non-SSO	• No exclusion required; however, all cases of gastroenteritis or enteritis should be regarded as potentially infectious and should remain home from work, school or daycare until 48 hours after diarrhea has stopped. • Refer to their physician for assessment and testing, if required.
Asymptomatic	Non-SSO	• No exclusion – contacts should monitor themselves for gastrointestinal symptoms, maintain good hand hygiene and food handling practices and seek medical attention if symptoms develop.

Preventive Measures

- Educate the **public** about the following:
 - personal hygiene, especially the sanitary disposal of items containing feces,
 - careful hand washing after defecation and sexual contact, and before preparing or eating food,
 - washing cutting boards, counter tops and utensils with soap and water after contact with raw meat,
 - cooking meats thoroughly,
 - methods of personal protective measures, in particular the correct and consistent use of condoms and discuss risk of sexual practices that permit fecal-oral contact,
 - accessing and drinking safe water supplies, and
 - testing private water supplies for presence of parasitic contamination, if suspected.
- Educate **food handlers** about:
 - proper food and equipment handling, preparation, and hygiene, especially in avoiding cross-contamination from raw meat products, and
 - thorough hand washing before and after meal preparation.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
November 2021	General	- Updated Template - Etiology, Clinical Presentation, Diagnosis and Treatment sections moved to Epidemiology - Key Investigation section moved to Public Health Management (formerly called Control) - Updated web links
	Reporting	Added a bullet on outbreak reporting
	Epidemiology	Updated sections or added sections that were missing
	Management of Case and Contacts	- Specified where terminology is SSO and included information in tables - Updated Exclusions to align with decisions agreed to in Enteric Transmission Risk Assessment and Exclusion Table - Aligned wording with other enteric diseases
	Preventive Measures	Reworded to align with other enteric diseases

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